

# Multimodal Semantic Representation Learning for Disaster Retrieval

Learning a Shared Semantic Embedding Space  
for Cross-Event Retrieval

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GitHub Repository: <https://github.com/Johnson-Opadere/multimodal-disaster-representation-learning>

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# High-level System Architecture

**Goal:** Align RGB, SAR, Reports and Narrations into a shared semantic embedding space

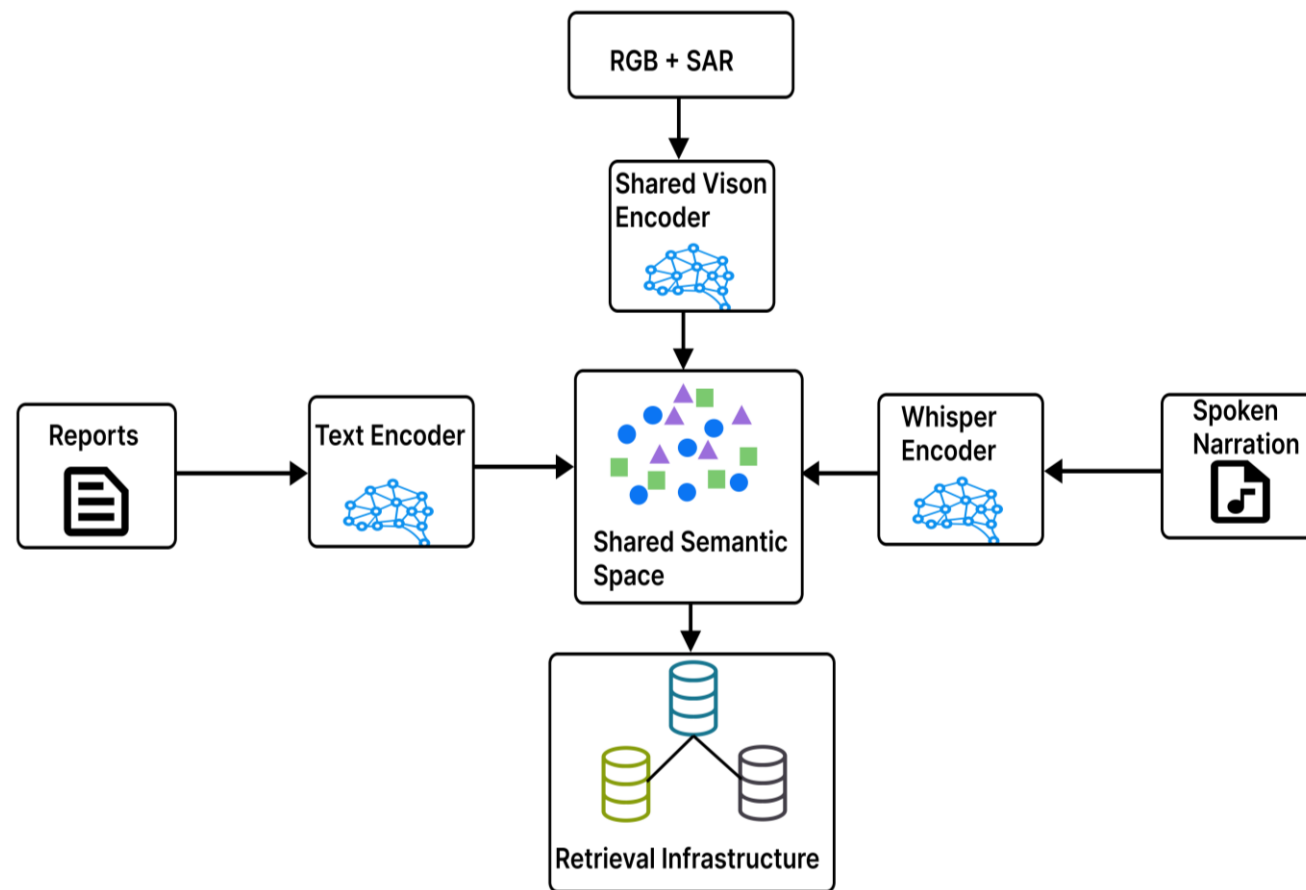
**Input:** RGB + SAR + Reports + Narration

**Method:** Contrastive Learning

**Output:** Shared Semantic Embedding Space

**Aligned Semantic Embeddings enable**

- Cross-modal retrieval
- Semantic search
- Nearest-neighbor retrieval
- Foundation for Project 2B



Enables Cross-Modal Retrieval and Semantic Search

# Why Retrieval Instead of Classification?

## Classification

Image → Classifier → “What class is this?”

[Flood, Wildfire, Earthquake, Tsunami]

### Limitations:

- Fixed labels
- No evidence retrievals
- Limited reasoning
- Poor for RAG

## Multimodal Retrieval

Image → Retrieval → “What semantically similar disasters exist?”

[Flood reports, Flood Narration, Flood Imagery, Cross-event evidence]

### Benefits:

- Multimodal search
- Evidence retrieval
- Cross-event generalization
- Foundation for RAG

## Why retrieval matters

Not simply what disaster is this, but what semantically related evidence exists across all modalities?

Project 2A reframes disaster understanding as multimodal semantic retrieval rather than closed-set classification

# Multimodal Inputs

Combining complementary visual and language modalities for disaster understanding

## RGB



- Appearance
- Texture
- Scene context

### Examples:

- Flooded roads
- Burned vegetations
- Collapsed buildings

## SAR



- Structural change
- Radar backscatter
- All-weather sensing

### Examples:

- Surface deformation
- Flood extent
- Infrastructure damage

## REPORTS



- Formal semantics
- Damage descriptions
- Situational context

### Example:

“Thousands displaced by flooding...”

## NARRATIONS



- Spoken semantics
- Eyewitness accounts
- Operational updates

### Example:

“Entire neighborhoods remain underwater...”

RGB + SAR + Language = Richer Disaster Understanding

Each modality contributes complementary information that no single modality can capture alone.

# Change-Aware Vision Encoder

## Learning semantic disaster representations from RGB and SAR imagery

### ResNet18 Backbone (pre-trained on ImageNet)

- Captures appearance, texture, and scene evidence

### Lightweight CNN

- Captures structural change and radar-specific patterns

### Temporal Differencing

- Focus is on disaster induced change rather than static appearance

### Multimodal Fusion

- Concatenation + Fusion MLP

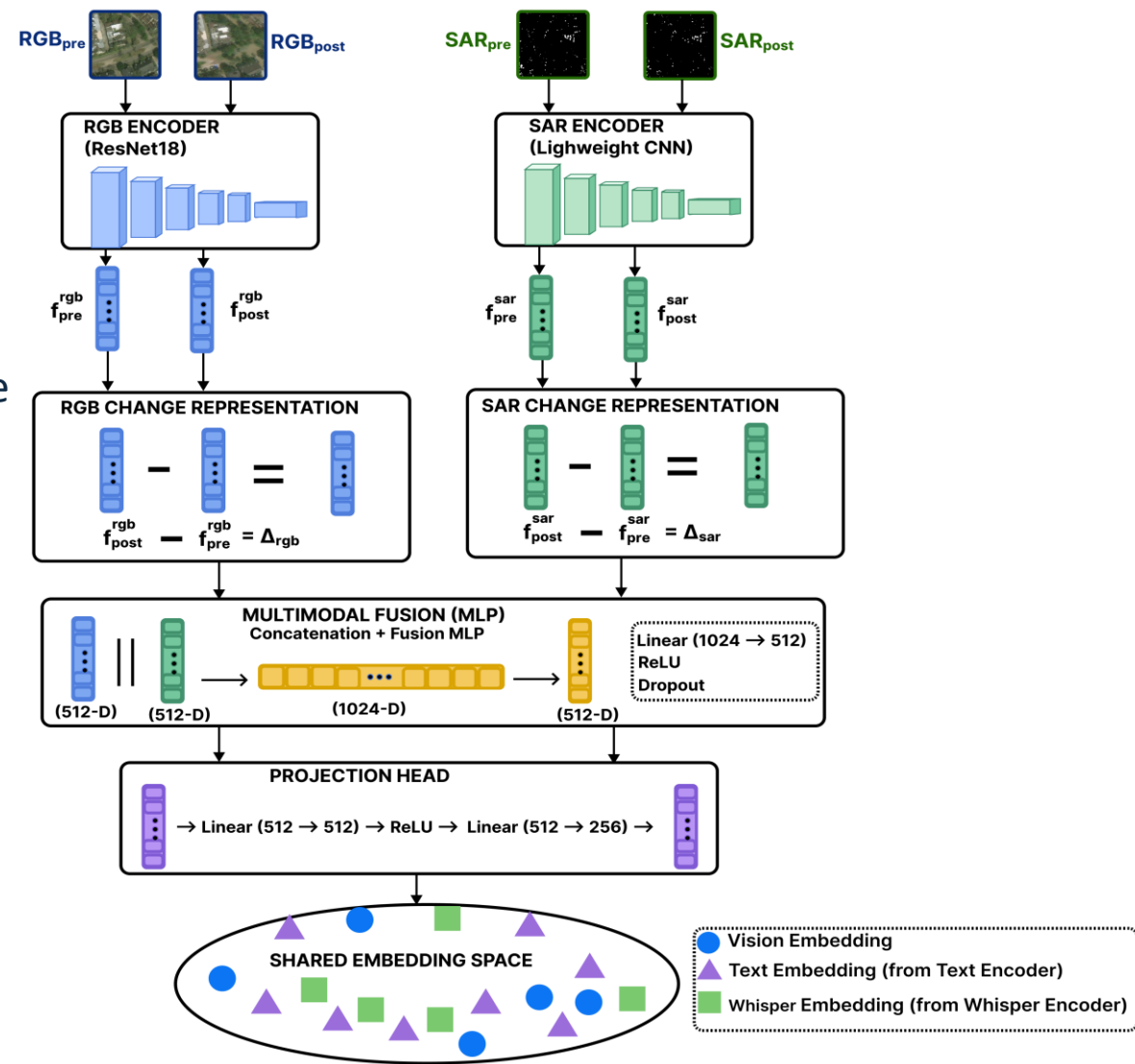
### Projection Head

- Map fused features into a retrieval-oriented embedding space

### Final Vision Embedding

- $L_2$  normalized  $z_{vision} \in R^{256}$

The Vision Encoder transforms RGB and SAR change signals into a unified semantic embedding used for multimodal retrieval



# Multimodal Contrastive Training Pipeline

Learning a shared semantic embedding space across vision, reports, and narrations

## SemanticBatchSampler

- Balances semantic batches

## Multi-positive InfoNCE

- Multiple positives per anchor

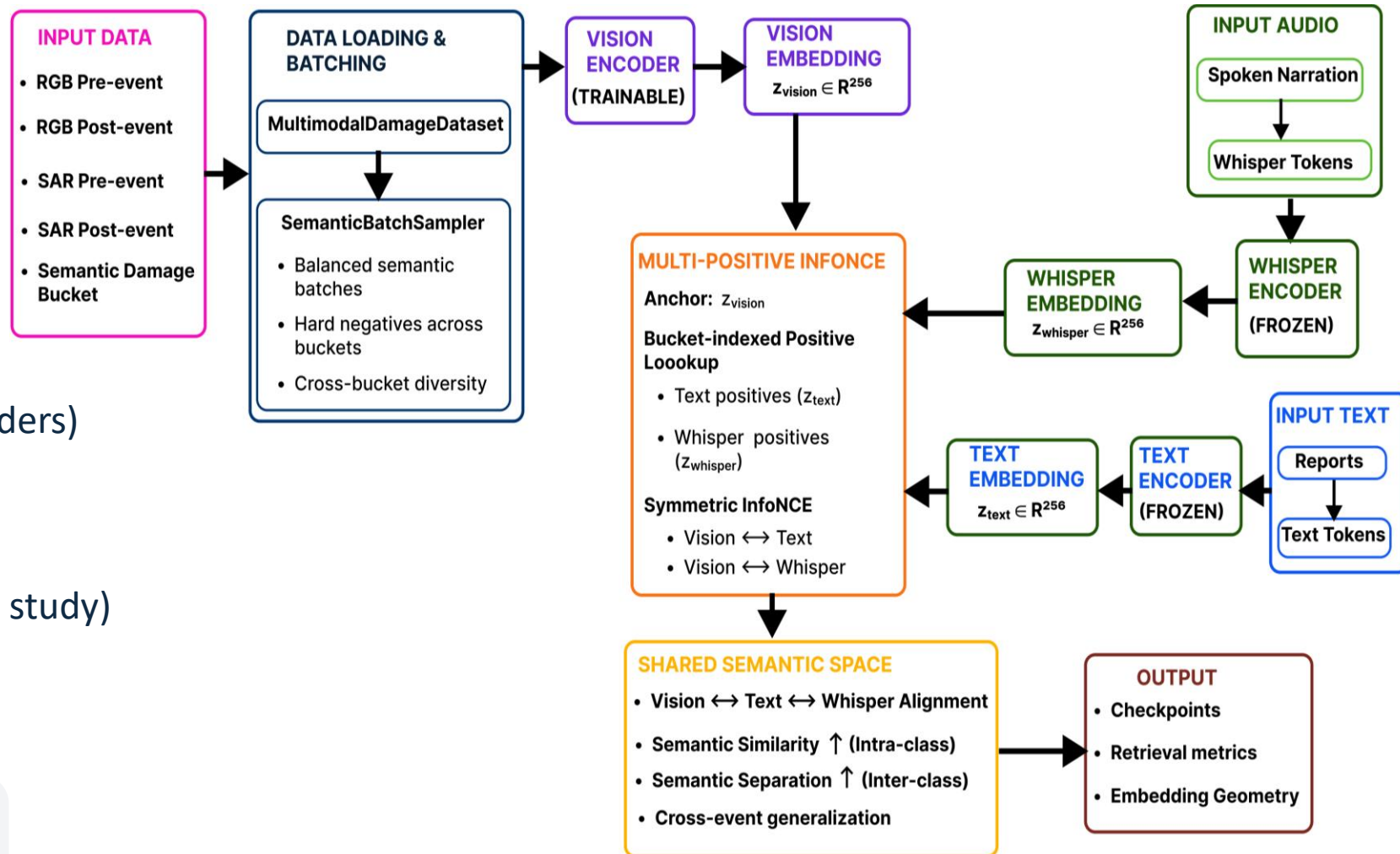
## Frozen DistilBERT (Text and Whisper Encoders)

- Preserve language semantics

## Layer4 Finetuning (selected from ablation study)

- Best adaptation/stability tradeoff

Project 2A learns multimodal semantic alignment through balanced semantic batches, multi-positive contrastive supervision, and selective vision finetuning.



# Multi-positive Contrastive Learning

Aligning vision, reports, and narrations through semantic neighborhoods

## Vision Anchor

- $Z_{vision}$

## Multiple positives

- Reports + Narrations from same bucket

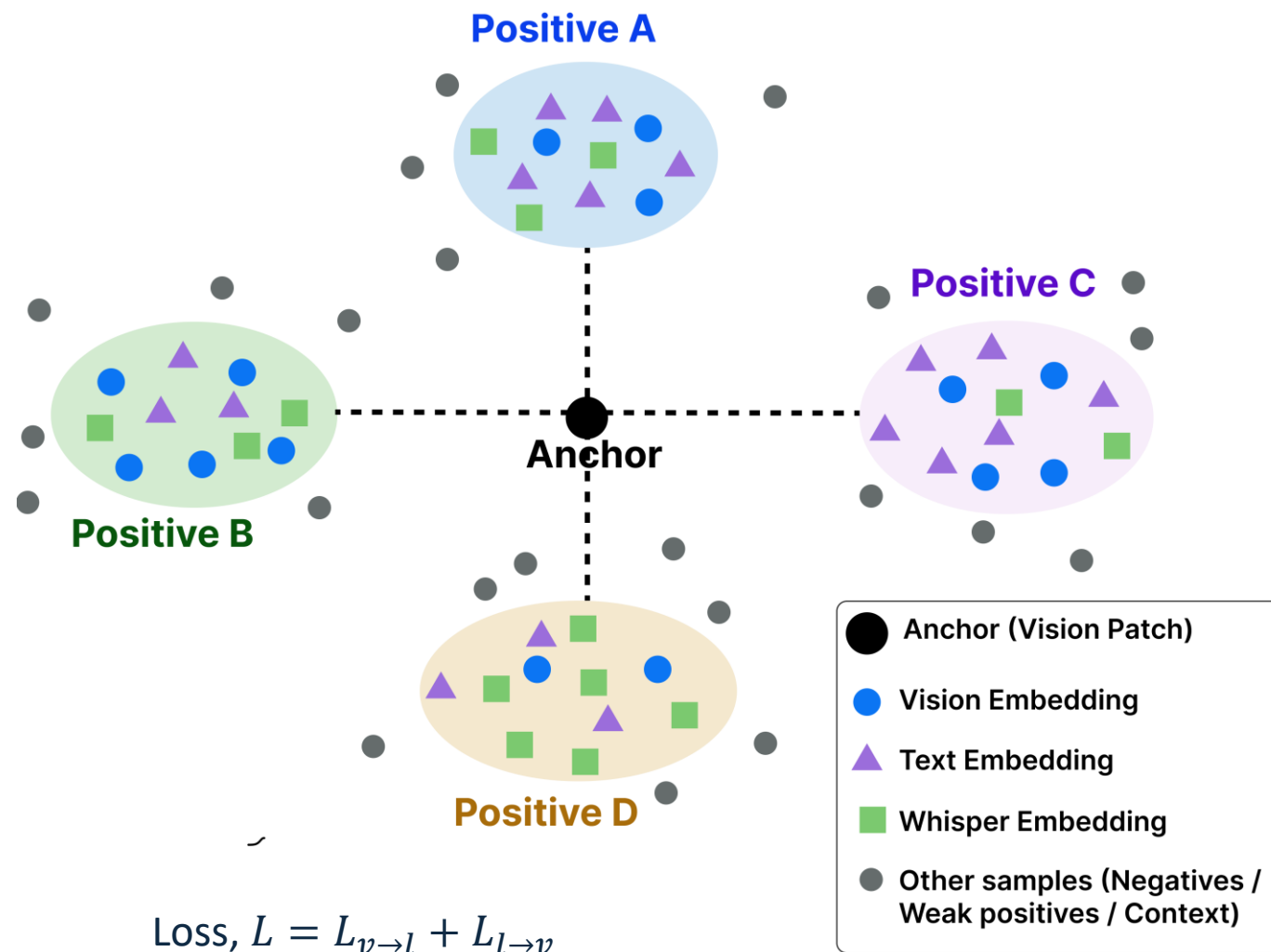
## Cross-Modal Supervision (Alignment)

- Vision  $\leftrightarrow$  Text
- Vision  $\leftrightarrow$  Whisper

## Symmetric InfoNCE

- Vision  $\leftrightarrow$  Language
- Language  $\leftrightarrow$  Vision

Multi-positive supervision produces richer semantic neighborhoods and more robust retrieval geometry.



# Learned Semantic Geometry

## Visualizing the shared multimodal embedding space

### Semantic clusters

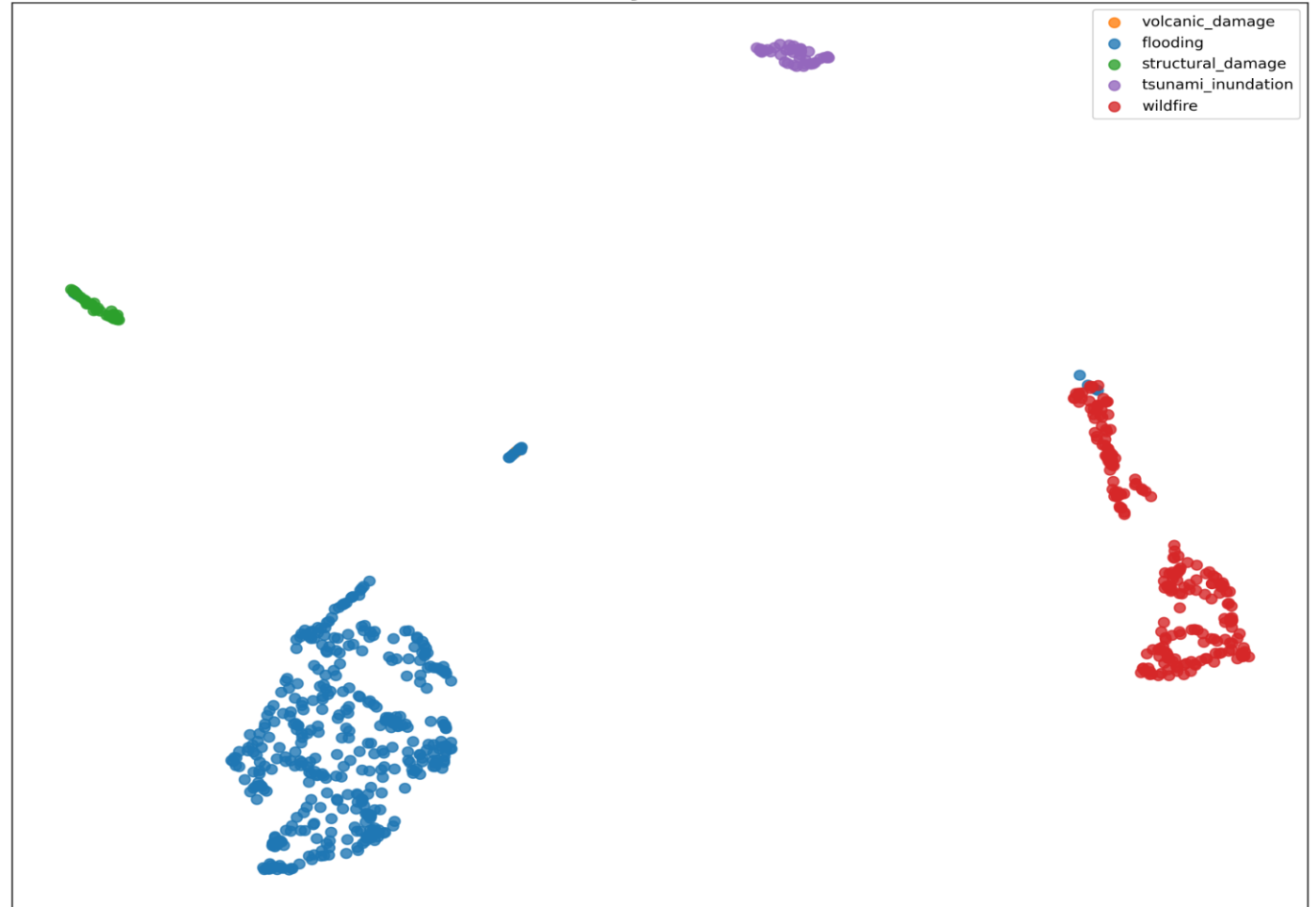
- Flooding
- Wildfire
- Structural Damage

### Retrieval-oriented geometry

- Clear inter-class separation
- Non-collapsing embedding space
- Semantically meaningful neighborhoods

Clear semantic clustering indicates successful representation learning for downstream retrieval.

UMAP Semantic Clusters (Vision)  
Project 2A





# Training Strategy and Ablations

Selecting the final representation learning configuration

## Finetuning Strategy

**Frozen:** No adaptation

**Layer4:** High-level adaptation

**Full:** End-to-end finetuning

★ **Layer4** selected

Reason: Best adaptation / stability tradeoff

## Positive Supervision

**Single Positive:** One positive per anchor

**Multi-Positive:** Multiple positives per anchor

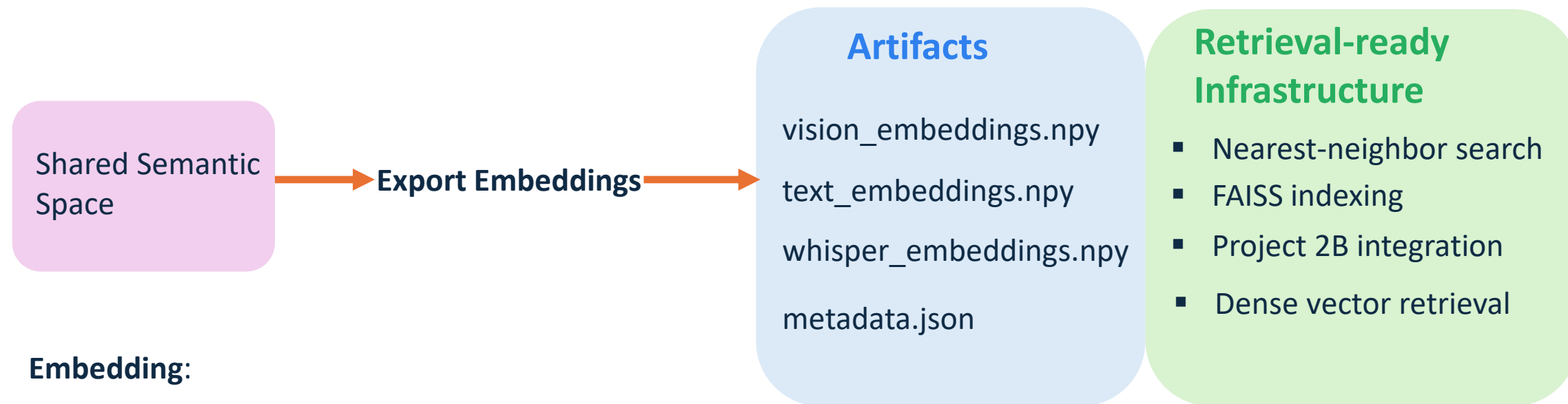
★ **Multi-Positive** selected

Reason: Richer semantic neighborhoods

Layer4 finetuning and multi-positive supervision provided the strongest retrieval-oriented representations.

# Exported Retrieval Infrastructure

Training outputs converted to retrieval-ready assets



## Embedding:

- 256-D
- $L_2$  Normalized
- Cosine similarity ready

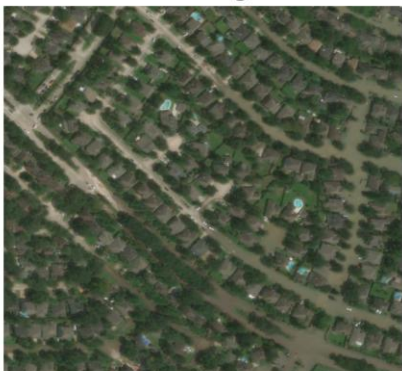
Project 2A produces retrieval-ready embeddings that serve as the foundation for Project 2B.

# Cross-Event Retrieval Results

Nearest neighbors emerge from learned disaster semantics rather than event identity

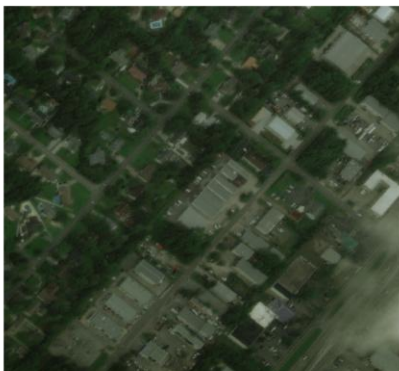
QUERY

hurricane-harvey  
flooding



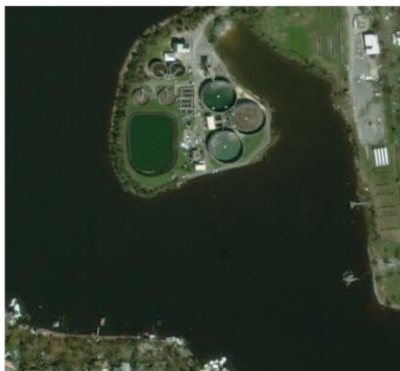
Rank #1

hurricane-florence  
flooding  
sim=0.986



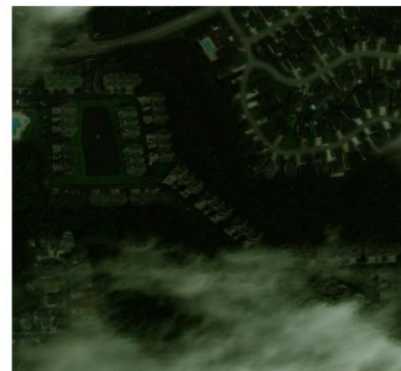
Rank #2

hurricane-michael  
flooding  
sim=0.981



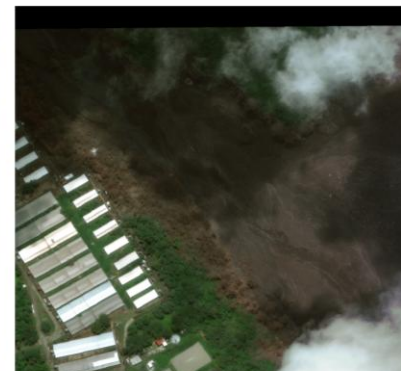
Rank #3

hurricane-florence  
flooding  
sim=0.911



Rank #4

guatemala-volcano  
volcanic\_damage  
sim=0.901



Rank #5

hurricane-florence  
flooding  
sim=0.895



QUERY

santa-rosa-wildfire  
wildfire



Rank #1

socal-fire  
wildfire  
sim=0.997



Rank #2

socal-fire  
wildfire  
sim=0.995



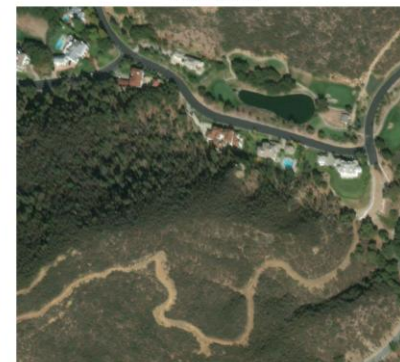
Rank #3

socal-fire  
wildfire  
sim=0.994



Rank #4

socal-fire  
wildfire  
sim=0.993



Rank #5

socal-fire  
wildfire  
sim=0.993

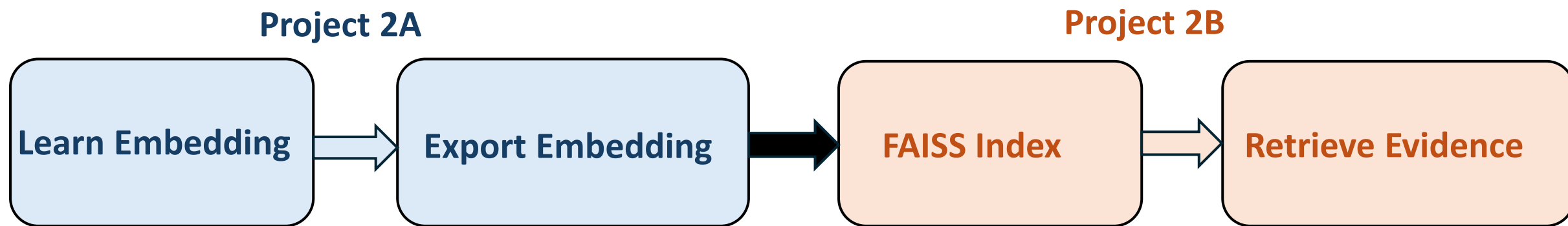


## Key observations:

- Query event is excluded from retrieval
- Retrieved neighbors originated from different events
- Flooding retrieves flooding; wildfire retrieves wildfire
- Embeddings support cross-event semantic search

# Representation Learning → Retrieval Infrastructure

Project 2A provides the semantic foundation for project 2B



## Project 2A

- Representation Learning
- Contrastive Learning
- Embedding Geometry
- Semantic Alignment

## Project 2B

- Retrieval Systems
- FAISS Search
- Retrieval Logic
- Evidence Retrieval

Project 2A produces the retrieval-ready embeddings that power Project 2B's multimodal search pipeline.

# Project 2A Summary

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- ✓ **Multimodal Retrieval**
- ✓ **RGB + SAR Change Representation Learning**
- ✓ **Multi-Positive Contrastive Learning**
- ✓ **Shared Semantic Embedding Space**
- ✓ **Cross-Event Generalization**
- ✓ **Retrieval-Ready Infrastructure**

Project 2A produces the retrieval-ready embeddings that power Project 2B's multimodal search pipeline.

**Foundation for Project 2B:  
Multimodal Retrieval & RAG Infrastructure**